PSYCHOLOGY IN GAME DESIGN

COUNTER-STRIKE 2

A Tale of Two Economies

How a genre-defining competitive shooter combines elegant design with a monetisation system shaped by gambling psychology.

PRESENTER

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PROGRAMME

MSc Psychology in Game Design and Digital Innovation

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Hours

750 – 1000 Hours

MODULE: PSYM179

GAME OVERVIEW

Counter-Strike 2: Scope and Significance

Verified figures drawn from Valve's official CS2 site and SteamDB charts captured on 21 April 2026.

ALL-TIME PEAK

1.86M

SteamDB reports 1,862,531 peak concurrent players.

MONTHLY PLAYERS

32.7M

Valve reports 32.7M monthly players through its official CS2 platforms.

FRANCHISE AGE

27Y

Counter-Strike has been a pillar of competitive gaming since 1999.

- **Valve, 2023:** a Source 2 successor to CS:GO within one of PC gaming's longest-running competitive franchises.
- **Tactical FPS:** the game prioritises precision, information management, positioning, and coordinated execution.
- **Target demographic:** primarily competitive players aged 16-30, though free access removes any effective age gate.
- **Steam user reviews:** Mostly positive (current rating as of April 2026).

Why this matters for analysis

CS2 is influential enough that its design choices matter beyond entertainment. The core question is not whether CS2 is successful, but what kinds of success its two economies are designed to produce.

TAKEAWAY CS2 IS NOT NICHE ENTERTAINMENT; IT IS A MASS, PERSISTENT COMPETITIVE PLATFORM WITH WIDE PLAYER ACCESS.

Core gameplay loop

Game of two halves

The game has 2 different sides, each with a maximum of 12 rounds per half. One side is defending the bomb and the second is trying to plant and get it to explode. First to 13 rounds wins.

Round by round

Each round starts with a buy phase, where players use their available money to purchase their loadouts for the upcoming round. The round starts and actions and events impact the total income, affecting the next rounds budget.

CORE GAMEPLAY DESIGN: PART II

Map Design: The Geometry of Skill

Sightline geometry on Dust 2: overlapping angles of engagement from key positions.

Spatial Reasoning

CS2 maps are designed as decision architectures. Success requires players to maintain ongoing situation awareness across multiple spatial variables: sightlines, rotation timings, and teammate positions.

Spatial Cognitive and Situation Awareness (Endsley, 1985)

Community callout map for Dust 2: a shared language built around spatial mastery.

Information Management

Maps dictate the flow of information. Skilled players construct and continuously update mental models of opponent positing based on fragmentary cues, turning spatial awareness into a competitive advantage.

Mental Models in Design (Norman, 1988)

From Experience: learning to read layouts and developing spatial awareness were the most significant skill transitions from casual to competitive play.

CORE GAMEPLAY DESIGN: PART III

Sound and Gunplay: The Mechanics of Mastery

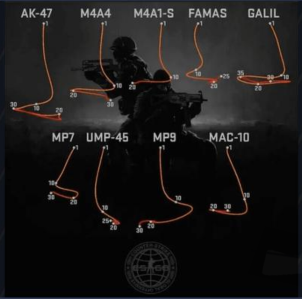
Sound Awareness

Sound is a primary, deterministic data source. Footsteps, utility cues, and weapon sounds provide precise information that skilled players must discriminate from ambient noise to predict enemy positions. The cost of a missed signal is death; the cost of a false alarm is positional disadvantage.

Signal Detection Theory (Green & Swets, 1966)

Common player positions across competitive maps: sound cues originate from predictable locations.

Fixed recoil patterns for primary rifles: each weapon demands unique muscle memory.



Gunplay Mastery

Recoil patterns in CS2 are fixed, not random. Success requires deliberate practice to master muscle memory and execution under pressure. Multiple mastery pathways (crosshair precision vs spray control vs game sense) support player autonomy, a core driver of intrinsic motivation.

Deliberate Practice (Ericsson et al., 1993) ; Self-Determination Theory (Ryan & Deci, 2000)

The Result: A system where intrinsic motivation is driven by the mastery of complex, predictable systems.

THE FIRST ECONOMY : IN-GAME

The Strategic Depth of Resource Management

Save

Preserve current weapons by avoiding risky engagements. Resources are protected rather than spent. No money spent at all.

Eco

Minimal spending on low-cost utility or pistols to maintain a viable future economy.

Force

Spending all available resources on sub-optimal gear to create a high-risk opportunity.

Full Buy

Maximum investments in rifles, armour, and full utility for peak competitive strength.

loop where every Round outcomes directly determine the next round’s budget, creating a recursive economic decision has both immediate combat value and downstream consequences.

The Scoreboard: Tracking team economy and loss bonuses for strategic planning.

First introduced in Counter-strike (1999), this economy system became the foundational template for the tactical FPS genre. Decisions to buy, save, or force-buy create a layer of strategic depth that rewards long-term planning and team coordination.

The Result: A system where strategic judgment is as important as physical execution.

THE FIRST ECONOMY - ECONOMIC PSYCHOLOGY

Economic Decisions Become Psychological Decisions

Prospect Theory

Players' reference points shift is based on their economic position. When facing likely losses, teams become risk-seeking, choosing force-buys with negative expected value. When ahead, they become risk-adverse, protecting their lead. This asymmetry is the core prediction of prospect theory.

Prospect Theory (Kahneman & Tversky, 1979)

Framing Effects

The same economic total can be read as separate scarcity or strategic investment. Interpretation shapes decision-making as much as the number itself.

Framing Effect (Tversky & Kahneman, 1981)

Cognitive load

During the buy phase, players simultaneously assess their own economy, estimate opponent economy, evaluate purchases, and coordinate with teammates, all within a time limit.

Cognitive Load Theory (Sweller, 1988)

Social Dilemmas

The economy forces teams to coordinate. Individual temptation to buy conflicts with collective optimal strategy, creating a classic social dilemma where personal preferences and team success diverge.

Social Dilemmas (Dawes, 1980)

From experience: economy disagreements are one of the most common sources of in-game conflict, often more divisive than individual mastery.

The Pivot: This is an economy on skill, strategy, and coordination. CS2 also contains a second economy, and it operates on very different psychological principles.

THE SECOND ECONOMY : OUT-OF-GAME

Case Openings Use Gambling Psychology

The reward has no gameplay impact, but the opening sequence is engineered to sustain anticipation, repetition, and perceived value.

01 - VARIABLE REINFORCEMENT

Case openings use a variable ratio schedule. The unpredictability of the reward creates a powerful psychological hook that encourages repetition.

02 - NEAR-MISS EFFECT

The visual presentation scrolls rare items visibly past the selector, manufacturing the feeling of almost winning. Critically, the outcome is determined server-side before the animation begins. **The near-misses are entirely fabricated**, a deliberate design choice.

03 – CLASSICAL CONDITIONING

Sound, pacing, colour glow, and reveal timing become **conditioned cues for arousal**. Over repeated openings, the animation itself produces anticipatory excitement before the item is revealed (Dixon et al., 2014).

04 - ANCHORING

The existence of skins worth thousands or more makes each low-cost opening feel connected to an extraordinary payoff. **The high-end market anchors perceived value for every participant.**

Operant Conditioning (Skinner, 1953)

A community-designed weapon skin in competitive play: cosmetic value meets gameplay context

Near-Miss Effect (Reid, 1986)

The case-opening ecosystem exists alongside, and profits from, the competitive experience.

Image presenting how potential high value outcomes provides justification for the relatively low investment price.

THE RESULT A system where **psychological conditioning** is used to drive engagement and spend, creating a uniquely demanding monetisation experience.

THE SECOND ECONOMY : LENS I & II

A £1.99 Opening Points Toward Million-Dollar Outcomes

ENTRY POINT

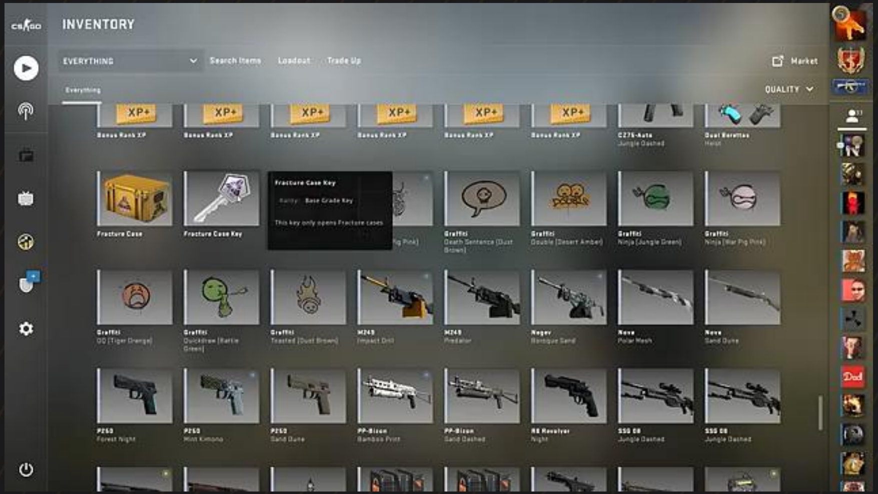
£1.99

A £1.99 key plus the market price of the case : a low-friction entry to a high-stakes system.

HIGH-END OUTCOME

$1M+

SteamAnalyst (2026) records an AK-47 Case Hardened #661 (StatTrak, Factory New, 4x Titan Holo) at approximately $1,000,000 + in a confirmed sale.



Case-opening animation (video): the spinning interface that drives the near-miss experience.

(Key prices shown in GBP; high-value sales typically reported in USD.)

Zendle and Cairns (2018) found a significant relationship between loot box spending and problem gambling severity.

Drummond and Sauer (2018) found that many loot box systems meet the psychological criteria for gambling.

SKIN MARKETPLACE – LENS III

Skins Created a Parallel Financial Economy

Once cases generate items, those items move into a market where rarity, visibility, and speculation create real monetary stakes.

01 - MARKET INFRASTRUCTURE

CS2 skins circulate through the Steam Community Market and third-party trading platforms, turning cosmetic drops into assets that can be listed, priced, and exchanged for real money.

$3.8 BILLION+

ESTIMATED ANNUAL MARKET VOLUME
(SteamAnalyst, 2026)

Personal reflection: the marketplace transforms how you perceive in-game items. A weapon skin is no longer just cosmetic; it has a price tag you are always aware of.

02 - HOW VALUE IS PRODUCED

Skin prices are shaped by technical and symbolic variables that make some items dramatically more desirable than others.

FLOAT VALUE
Wear is measured precisely, so small numerical differences can produce major price gaps.

PATTERN INDEX
Specific visual variants create collector hierarchies within a single skin type.

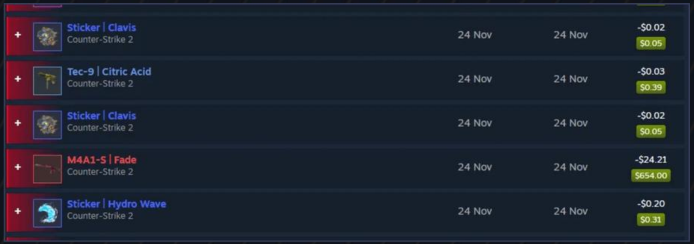
STICKERS AND PROVENANCE
Applied stickers, tournament history, and cultural prestige can multiply market value.

03 - BEHAVIOURAL ECONOMICS

ENDOWMENT EFFECT
Ownership makes players overvalue rare items relative to the wider market. (Thaler, 1980)

SUNK COST FALLACY
Money already spent keeps players attached to the ecosystem even when enjoyment declines. (Arkes & Blumer, 1985)

LOSS AVERSION
Selling below a purchase price feels like a meaningful loss, sustaining inflated expectations. (Kahneman & Tversky, 1979)



Steam Community Market transaction history: real-money price movements on individual skin trades.

local prices in £, global market figures in $

KEY QUESTION

If minors can access a system with **real-money trading, speculation, and information asymmetry**, at what point does a game feature become a financial product?

ETHICAL IMPLICATIONS

The Ethical Problem Is Not Abstract

2016

CSGOLotto made the issue visible

Syndicate and TmarTn promoted a skin-gambling site without disclosing ownership. The platform was presented as exciting and trustworthy to viewers that included minors.

(Federal Trade Commission, 2017)

SCALE

The scandal was part of a wider ecosystem

CSGOLotto was not an isolated case. Sites such as CSGOLounge and CSGOJackpot turned tradeable skins into gambling currency through the wider Steam trading infrastructure.

Valve

The response was reactive, not preventive

Valve issued cease-and-desist letters only after the ecosystem had already grown in public view. The company enabled the infrastructure and benefited from marketplace activity.

PERSONAL POSITION

I was part of the target audience

I watched Syndicate and TmarTn during this period. The gambling ecosystem was not hidden from me; it was presented as entertainment.

CSGOLotto: the skin-gambling platform at the centre of the 2016 disclosure scandal.

A system can be harmful before it is formally prohibited. When young players access gambling-like mechanics without meaningful safeguards, the ethical problem exceeds simple legal compliance.

TAKEAWAY **Did valve fail to foresee these consequences, or did it benefit from not acting until the damage was visible?**

REGULATORY RESPONSE

Regulation Lags Behind the Psychology

01 - BELGIUM

The gaming commission moved toward prohibition, ruling that loot boxes are games of chance. This is one of the clearest regulatory interventions in Europe, prioritising consumer protection over industry freedom.

(Belgian Gaming Commission, 2018)

02 - NETHERLANDS

The Gaming Authority initially took a similar position, though enforcement was subsequently narrowed by court ruling. The focus remains on tradeable value and the potential for real-money conversion.

(Kansspelautoriteit, 2018)

03 - AUSTRALIA

A senate inquiry recommended that loot boxes be restricted to adults, citing the psychological similarities between these systems and traditional gambling, particularly for vulnerable audiences.

(Parliament of Australia, 2018)

04 - UNITED KINGDOM

The Gambling Commission has called for industry self-regulation while acknowledging the risks for younger players but has not yet reclassified loot boxes as gambling under current law.

(UK Gambling Commission, 2017)

Zendle and Cairns (2018) found that loot box spending is significantly linked to problem gambling severity, reinforcing that psychological evidence exists even where regulation has not caught up.

Closing Question: If a system produces the same behavioural responses as gambling, how much should the legal label matter?

CONCLUSION

Two Economies, One company

01 - ECONOMY ONE: SKILL

A genre-defining mechanic that can be seen across all tactical shooters now. It rewards planning, coordination, and mastery, defining the competitive experience.

02 - ECONOMY TWO: MONETISATION

A system that puts financial incentive over the safety of users, including children. It leverages gambling psychology to drive engagement and speculative value.

The question for the industry is whether these **two design philosophies** can coexist responsibly.

The **same** company designed **both.**

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Source note: Player-count figures in the deck were taken from Valve's official CS2 site and SteamDB. High-value skin figures were phrased conservatively to distinguish documented sales from estimated valuations and rejected offers.